Technische Hochschule Ingolstadt

Exposé

in the [DEGREE PROGRAM]
Faculty of Computer Science

Thesis Title

[YOUR THESIS TITLE HERE]

First and Last Name: **[Your Full Name]**

Student ID: [XXXXXXXX]

First Supervisor: Prof. Dr. [Supervisor Name]

Second Supervisor: [TBD]

Submission Date: [DD.MM.YYYY]

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1 Introduction and Motivation

Purpose of this section: *Introduce the topic to a reader who is knowledgeable in the field but unfamiliar with your specific problem. Aim for 1–2 pages.*

What to include:

- ***Context & relevance*** *— Why does this topic matter today? Mention industry trends, societal challenges, or scientific gaps that make this research timely.*
- ***Problem hook*** *— Describe a concrete situation or deficiency that your work will address. Use statistics or a recent event if available.*
- ***Scope statement*** *— Briefly state what the thesis will and will* not *cover.*
- ***Structure overview*** *— One sentence per subsequent section explaining how the exposé is organised.*

Tip: *Write this section last, once all other sections are complete.*

2 Problem Statement and Research Questions

Purpose of this section: *Precisely articulate the gap, challenge, or open question your thesis addresses. Aim for 0.5–1 page.*

What to include:

- *A concise statement (2–4 sentences) of the core problem.*
- *An explanation of* why *existing solutions are insufficient.*
- ***Research questions (RQs)*** *— formulate 1–3 specific, answerable questions. Each RQ should be:*
  - Specific*: addresses one concept at a time.*
  - Feasible*: answerable within the thesis timeline.*
  - Relevant*: directly linked to the problem.*
- *Optionally, a research* ***hypothesis*** *if you have one.*

Tip: *Your RQs are the anchor for every other section — methodology, evaluation criteria, and conclusions all flow from them.*

RQ1 [Write your first research question here.]

RQ2 [Write your second research question here.]

RQ3 [Optional: write a third research question, or delete this row.]

3 Objectives

Purpose of this section: *Translate your research questions into concrete, measurable goals. Aim for 0.5 pages.*

What to include:

- ***Primary objective*** *— the single most important goal (maps directly to your main RQ).*
- ***Secondary objectives*** *— supporting goals (e.g., design an artefact, conduct a user study, develop a framework).*
- *Each objective should start with an action verb:* design, implement, evaluate, analyse, compare, develop, validate, . . .

Tip: *Objectives are* not *research questions. An objective describes what you will* do*; an RQ asks what you want to* find out*.*

4 Related Work

Purpose of this section: *Demonstrate awareness of existing literature and position your work relative to it. Aim for 1–2 pages.*

What to include:

- ***Search strategy*** *(brief) — which databases, keywords, and date range you used (e.g., ACM Digital Library, IEEE Xplore, Google Scholar; last 10 years).*
- ***Thematic clusters*** *— group related papers by sub-topic, not by chronology. Use one subsection per cluster.*
- ***Critical synthesis*** *— do not merely summarise each paper; compare, contrast, and identify common limitations.*
- ***Gap identification*** *— conclude with the specific gap your thesis fills, explicitly referencing your RQs.*

Tip: *Every claim should be backed by a citation [?]. Aim for at least 10–15 sources in the exposé.*

4.1 [Topic Cluster 1 — e.g., Existing Approaches]

Summarise 3–5 papers that address the problem from one angle. Highlight their strengths and shared limitations.

4.2 [Topic Cluster 2 — e.g., Relevant Techniques / Frameworks]

Summarise 3–5 papers that provide methodological or technical foundations relevant to your work.

4.3 [Topic Cluster 3 — e.g., Gaps & Open Challenges]

Synthesise the overall state of the art and clearly identify the gap your thesis addresses. This subsection is the bridge to your problem statement.

5 Methodology

Purpose of this section: *Describe* how *you will answer your research questions. This is often the most critical section for supervisors. Aim for 1–2 pages.*

What to include:

- ***Research design*** *— qualitative, quantitative, or mixed? Empirical study, design science, systematic review, simulation, . . . ?*
- ***Use case / study context*** *— the specific system, domain, or dataset you will work with.*
- ***Variables or evaluation criteria*** *— what you measure and how (independent, dependent, control variables for experiments; or evaluation criteria for design artefacts).*
- ***Data collection*** *— surveys, interviews, logs, benchmarks, public datasets, . . .*
- ***Analysis methods*** *— statistical tests, qualitative coding, performance benchmarks, . . .*
- ***Validity considerations*** *— how you will ensure internal and external validity, reliability, and ethical compliance.*

Tip: *Justify every methodological choice by linking it back to your RQs.*

5.1 Use Case Definition

Describe the application domain, system, or dataset that will serve as the object of study. Explain why this use case is appropriate for answering your RQs. Include any relevant constraints (access, size, language, . . .).

5.2 Independent Variables

List and define the variables (or design parameters) that you will deliberately manipulate or vary. For each variable, state its possible values or levels.

5.3 Dependent Variables

List and define the outcomes you will measure. For each dependent variable, specify the measurement instrument or metric (e.g., task completion time in seconds, SUS score 0–100, F1-score, . . .).

5.4 Procedure / Research Steps

Provide a step-by-step overview of how the study or development work will be carried out — from setup and data collection through to analysis. A numbered list or small diagram works well here.

6 Expected Contributions and Results

Purpose of this section: *Communicate the anticipated value of your thesis. Aim for 0.5–1 page.*

What to include:

- ***Scientific contribution*** *— new knowledge, a validated model, a benchmark, an empirically tested framework, . . .*
- ***Practical / applied contribution*** *— a prototype, tool, guideline, or design recommendation.*
- ***Expected findings*** *— your educated hypothesis about what you will discover (be honest about uncertainty; phrases like* "we expect that . . ." *or* "it is hypothesised that . . ." *are appropriate).*
- ***Limitations*** *— acknowledge foreseeable constraints (sample size, generalisation, time, . . .).*

Tip: *Do not over-promise. Reviewers value intellectual honesty over grand claims.*

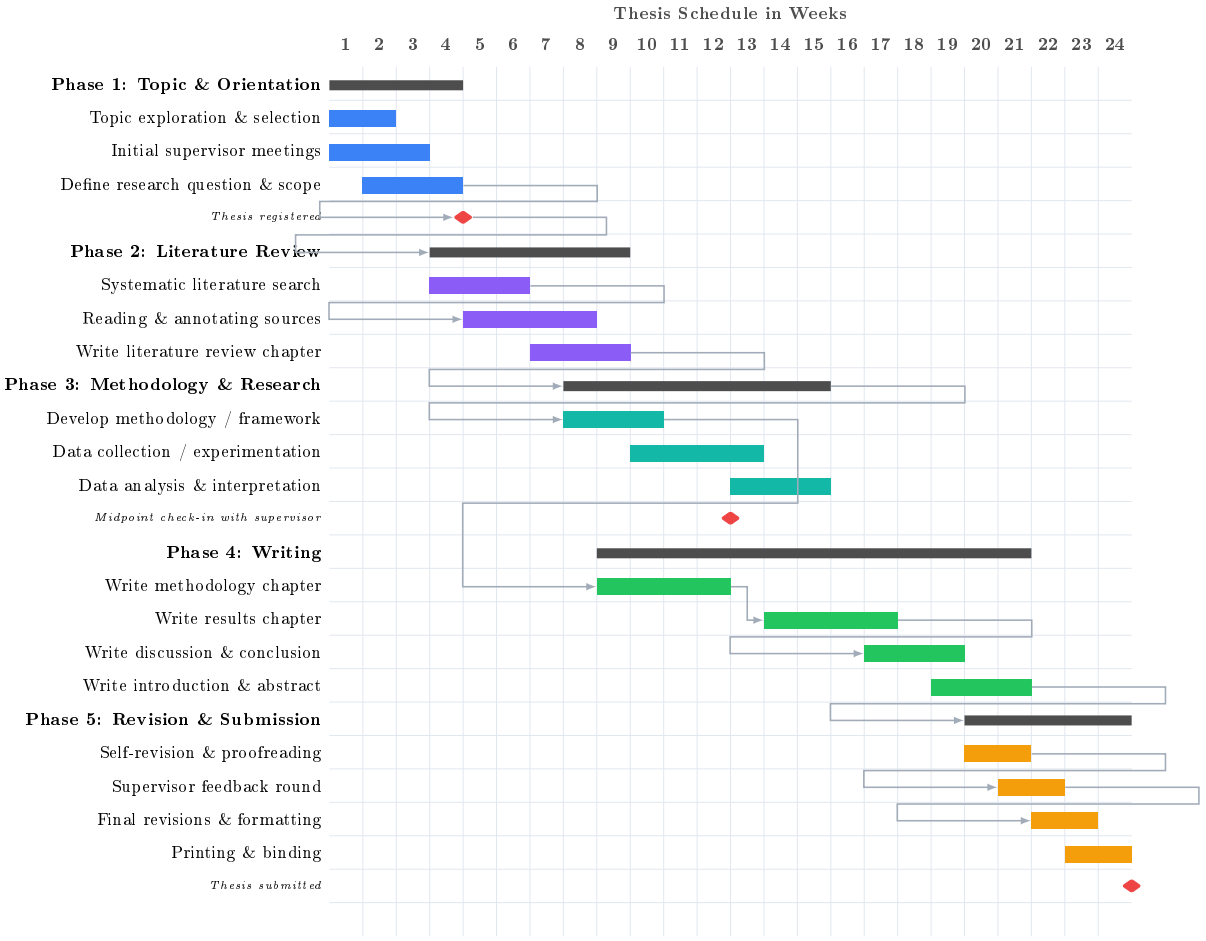
7 Work Plan and Schedule

Purpose of this section: *Demonstrate that the project is feasible within the allotted time. The Gantt chart below covers a typical 24-week (6-month) thesis period — adjust dates and tasks to match your actual timeline.*

What to include:

- *A Gantt chart (provided) showing phases, tasks, and milestones.*
- *A short paragraph (3–5 sentences) summarising the phases and any critical-path dependencies.*
- *Key milestones: registration, mid-point check-in, submission.*

Tip: *Build in buffer time before the final submission — unforeseen delays (data collection issues, feedback rounds) are the rule, not the exception.*


Thesis Schedule in Weeks
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Phase 1: Topic & Orientation
Topic exploration & selection
Initial supervisor meetings
Define research question & scope
Thesis registered
Phase 2: Literature Review
Systematic literature search
Reading & annotating sources
Write literature review chapter
Phase 3: Methodology & Research
Develop methodology / framework
Data collection / experimentation
Data analysis & interpretation
Midpoint check-in with supervisor
Phase 4: Writing
Write methodology chapter
Write results chapter
Write discussion & conclusion
Write introduction & abstract
Phase 5: Revision & Submission
Self-revision & proofreading
Supervisor feedback round
Final revisions & formatting
Printing & binding
Thesis submitted